SmartSupport AI

Agentic AI Customer Support Platform

Project Plan and Non-Functional Requirements

DEPI .NET Full Stack Track | Prepared by Khaled Osama

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1. Project Overview

SmartSupport AI is an intelligent customer support platform that uses Large Language Models and AI agents to handle customer inquiries 24/7. The system understands customer intent, retrieves answers from a knowledge base, performs actions such as checking order status and processing refunds, and hands complex cases over to human agents. It also provides analytics on AI performance and customer satisfaction.

Technology stack

ASP.NET Core MVC, Entity Framework Core, ASP.NET Identity, Azure OpenAI / OpenAI API, Semantic Kernel, SignalR, Azure Cognitive Search, Redis, Azure Blob Storage, Chart.js, Bootstrap 5, Microsoft Azure.

Design patterns

NTier Architecture, Repository, Unit of Work, Chain of Responsibility, Strategy, Observer, and Dependency Injection.

2. DEPI Deadlines

Item	Due date	Status
Project Planning and Management	16 Oct 2026	This document
Literature Review	16 Oct 2026	Separate document
Requirements Gathering	16 Oct 2026	Separate document
System Analysis and Design	6 Nov 2026	Planned
Implementation (source code and execution)	30 Nov 2026	Planned
Final Presentation, Testing and Reports	4 Dec 2026	Planned

3. Project Phases

Phase	Dates	Deadline	Main output
1. Planning, literature review, requirements	15 - 16 Oct	16 Oct	Project plan, literature review, requirements specification
2. System analysis and design	17 Oct - 6 Nov	6 Nov	Use cases, ERD, UML diagrams, NTier architecture, AI agent and prompt design, UI wireframes
3. Implementation	2 - 30 Nov	30 Nov	Working system deployed on Azure, with source code

Phase	Dates	Deadline	Main output
4. Testing, reports, presentation	23 Nov - 4 Dec	4 Dec	Test report, final documentation, presentation and demo

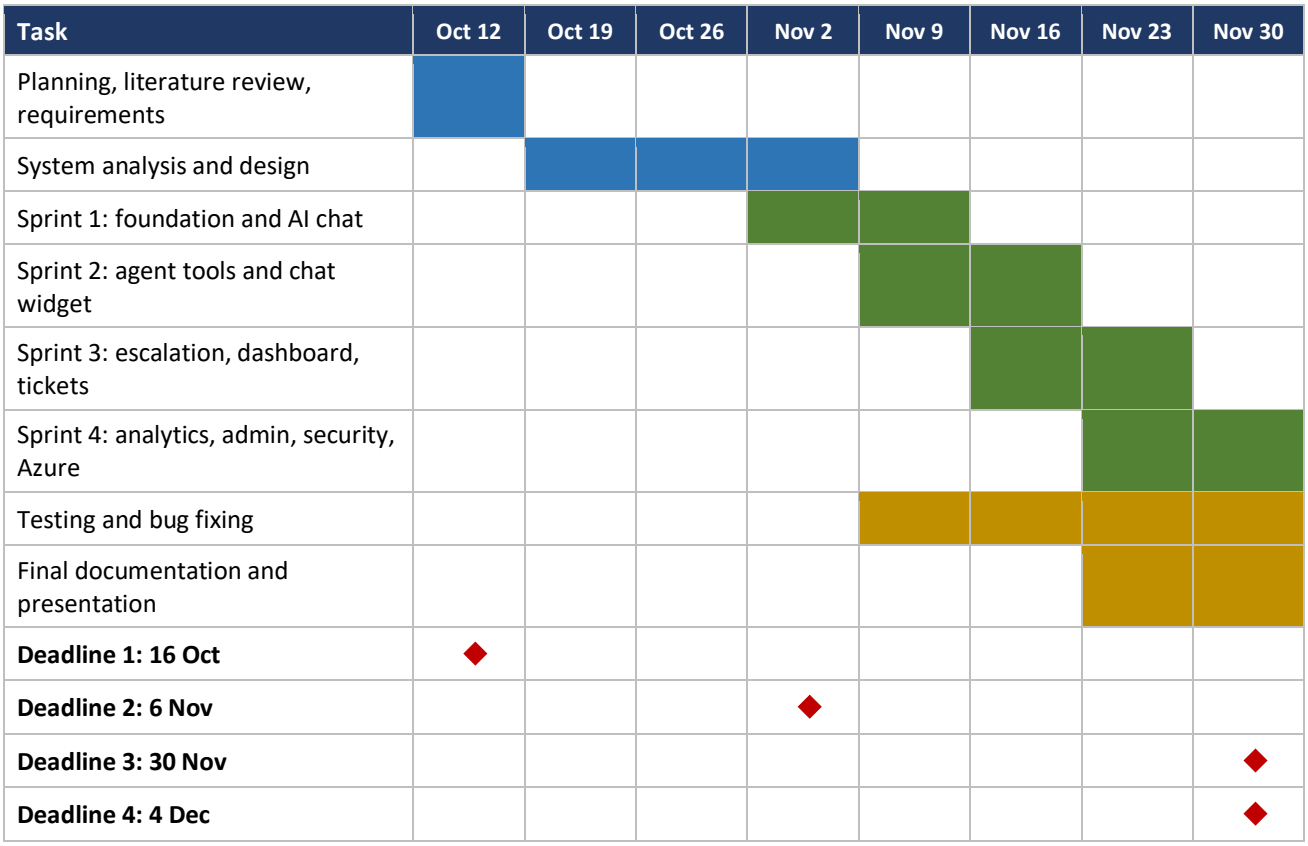
Phases 2 and 3 overlap in the week of 2 November: the code skeleton and Azure setup start while the design is being finalized.

Implementation sprints

Sprint	Dates	Scope
1	2 - 13 Nov	NTier solution (Web, Services, AI, Data), database, repositories and Unit of Work, ASP.NET Identity, Semantic Kernel setup, base AI chat, knowledge base with semantic search
2	9 - 20 Nov	Agent tools (order lookup, refund, callback, escalate, help articles), intent recognition and routing, SignalR chat widget with typing indicator
3	16 - 27 Nov	Escalation and AI-to-human handoff, agent dashboard with AI suggestions, ticket management and SLA tracking
4	23 - 30 Nov	Analytics with Chart.js, admin panel, PII handling and audit logging, Azure deployment, Redis caching, Application Insights

4. Gantt Chart

Weeks start on Monday. Blue: analysis and design. Green: implementation sprints. Gold: testing and documentation. The diamond marks a DEPI deadline.



5. Milestones

Date	Milestone	Exit criteria
16 Oct	Planning deliverables submitted	Project plan, literature review and requirements submitted to DEPI
6 Nov	Design package complete	ERD, architecture diagram, UML diagrams, prompt and agent design, UI wireframes
13 Nov	AI chat with knowledge base	Chat answers questions using semantic search over the knowledge base
20 Nov	Agent tools and widget demo-ready	Agent tools work; the chat widget runs in real time over SignalR
27 Nov	Escalation and tickets working	Escalation triggers fire; agent dashboard and ticket workflow function
30 Nov	Implementation submitted	Full system deployed on Azure with source code delivered
4 Dec	Final delivery	Test report, documentation and final presentation delivered

6. Scope Priority

- **Must have:** AI chat with knowledge base search; four agent tools (order lookup, refund, escalate, help articles); SignalR chat widget; escalation and agent dashboard; ticket management; basic analytics; Azure deployment.
- **Should have:** Intent routing (Chain of Responsibility and Strategy); sentiment analysis; SLA tracking; PII masking; audit log; Redis caching.
- **Could have (cut first):** A/B testing of responses; auto-suggested knowledge base updates; canned responses library; automatic language detection; extra conversation encryption.

7. Risks and Mitigation

Risk	Mitigation
Azure provisioning delays	Request Azure OpenAI and Search access during the design phase, not at implementation
Short implementation window (about 3.5 weeks)	Keep 'could have' items out of the plan unless Sprint 3 finishes early
Wrong AI actions, such as incorrect refunds	Require confirmation for money actions and log every AI action to the audit trail
LLM cost and latency	Cache common answers in Redis and limit the context size
Testing squeezed at the end	Write unit tests in each sprint; use 23 Nov - 4 Dec for integration testing and the report

8. Non-Functional Requirements

Category	Requirement
Performance	First AI token streams within 2 seconds; full reply within 8 seconds; cached answers within 500 ms
Scalability	Supports 200 concurrent chat sessions; SignalR scaled out through a Redis backplane

Category	Requirement
Availability	99.5% uptime on Azure App Service; chat falls back to ticket creation if the AI service is down
Security	ASP.NET Identity with role-based access (customer, agent, admin); HTTPS only; API keys stored in Azure Key Vault
Privacy and compliance	PII masked before reaching the LLM; conversations encrypted at rest; data retention policy
AI reliability	Answers grounded in the knowledge base; low confidence triggers escalation; every AI action audited
Maintainability	Layered NTier structure; AI services behind interfaces; unit tests on business and AI layers
Usability	Responsive on mobile with Bootstrap 5; typing indicator; language auto-detected
Observability	Application Insights tracks errors, latency and token usage